

Attempt any six (including subparts)

1 a) Write a tail-recursive algorithm (using C pseudocode) which takes the value of n from the user as input and displays the n^{th} term of the Fibonacci series as output. Specify its asymptotic time complexity (using smallest upper bound).

b) A complex-valued matrix X is represented by a pair of matrices (A, B) , where A and B contain real values. Write an algorithm (using C pseudocode) that computes the product of two complex-valued matrices (A, B) and (C, D) , where $(A, B) * (C, D) = (A+iB) * (C+iD) = (AC-BD) + i(AD+BC)$. Determine the asymptotic time complexity (using smallest upper bound) of your algorithm if the matrices are all of dimensions $n \times n$.

c) Explain with suitable examples 0-based indexing, 1-based indexing and n -based indexing for arrays.

(3+4+3)

2 Write down algorithms (using C pseudocode) to perform the following operations on a stack (implemented using array) of size n (where the value of n is provided by the user as input) and specify their asymptotic time complexities (using smallest upper bound).

- Push k number of elements (provided by the user as input) to the stack (where the value of k is also taken from the user as input)
- Reverse the elements of the given stack using recursion / the call stack only.
- Display the contents of the stack

(2+6+2)

3 a) Let an array be represented as $A(2:20, 3:30, 1:25, 0:25, 5:30)$. Assuming column major representation of the array (using sequential byte addressing scheme) with 2 bytes per element. If k is the address of $A(4,5,6,7,5)$ then the address of the element represented as $A(5,10,15,6,12)$ is _____ [Fill in the blank correctly].

b) For the following, find the smallest function $g(x)$ for which $f(x) = O(g(x))$

i) $f(x) = 2 + 6x + 18x^2/2! + 54x^3/3! \dots + \infty$

ii) $f(x) = 1 + x + x^2/2! + x^3/3! + \dots + x^n/n!$

- c) For the following function [where, $\text{end_index} - \text{start_index} = n-1$], using asymptotic analysis, the time complexity of `foo` (using smallest upper bound) is _____ and the recursion stack space required by `foo` (using smallest upper bound) is _____

```
int foo(int array[], int start_index, int end_index, int element){
    if (end_index >= start_index){
        int middle = start_index + (end_index - start_index) / 2;
        if (array[middle] == element)
            return middle;
        if (array[middle] > element)
            return foo(array, start_index, middle-1, element);
        return foo(array, middle+1, end_index, element);
    }
    return -1;
}
```

- d) Write an algorithm (using *C* pseudocode) to print the number of occurrences of each character in a given string (provided by the user as input) of length *n*. Determine the asymptotic time complexity (using smallest upper bound) of your algorithm.

(3+2+2+3)

- 4a) Write an algorithm (using *C* pseudocode) to perform the addition of two polynomials [where each polynomial is stored the array as a list of elements in the form (*m*, e_{m-1} , b_{m-1} , e_{m-2} , b_{m-2} , ..., e_0 , b_0) where, the first entry *m* represents the number of non-zero terms followed by two entries (e_i , b_i) for each non-zero term (representing an exponent-coefficient pair) in the order $e_{m-1} > e_{m-2} > \dots > e_0 \geq 0$] and print the resultant polynomial. Specify the asymptotic space and time complexity (using smallest upper bound) of your algorithm.

- b) Explain what is understood by a priority queue. What is the difference between direct and indirect recursion? What is tail recursion and what is its advantage?

- c) Prove that if $f(n) = a_m n^m + \dots + a_1 n + a_0$, the $f(n) = O(n^m)$, for $n \geq 1$, where $a_0, a_1, \dots, a_m$ are constants.

(4+3+3)

5. Write down algorithms (using C pseudocode) to perform the following operations on a non-circular queue (implemented using array) of size n (where the value of n is provided by the user as input) and specify their asymptotic time complexities (using smallest upper bound).

- Insert k number of elements (provided by the user as input) to the non-circular queue (where the value of k is also taken from the user as input).
- Reverse the elements of the non-circular queue using stack
- Display the contents of the circular queue

non

(2+6+2)

6 a) For the following function the best case asymptotic time complexity (using smallest upper bound) is _____ and the worst case asymptotic time complexity (using smallest upper bound) is _____

```
void myfunc(int array[], int size) {
    for (int step = 1; step < size; step++) { - size + 1
        int key = array[step];
        int j = step - 1;
        while (key < array[j] && j >= 0) {
            array[j + 1] = array[j];
            --j;
        }
        array[j + 1] = key;
    }
}
```

b) Write algorithms (using C pseudocode) for inserting and deleting an element (provided as input from the user) to the priority queue (implemented using array) of size n (where the value of n is provided by the user as input). Specify their asymptotic time complexities (using smallest upper bound).

c) Show whether the following are true or false with suitable proof/ justification.

- | | |
|---------------------------------|--|
| i) $6 \cdot 2^n + n^2 = O(2^n)$ | ii) $6 \cdot 2^n + n^2 = O(n^{10000})$ |
| iii) $10n^2 + 4n + 2 = O(2^n)$ | iv) $10n^2 + 4n + 2 = O(n^{2.5})$ |
| v) $10n^2 + 4n + 2 = O(n^2)$ | vi) $10n^2 + 4n + 2 = O(n)$ |
| vii) $n/500000 = O(500000)$ | viii) $500000 = O(5)$ |

(2+4+4)

- 7a) Write your recursive algorithms (using *C* pseudocode) to find the Greatest Common Divisor (GCD) and Least Common Multiple (LCM) of two given numbers (provided by the user as input). Specify their asymptotic time and space complexities (using smallest upper bound).
- b) State whether the statements are true or false:
- i) The computing time needed for multiply and divide operation generally depends of the numbers/operands involved in the operation.
 - ii) In Data Structure arrays are always implemented by using consecutive memory locations.
- c) Write non-recursive and non-tail recursive algorithms (using *C* pseudocode) to find the factorial of a given number (provided by the user as input). Specify their asymptotic time and space complexities (using smallest upper bound).

(4+2+4)

c) Write an algorithm (using C pseudocode) to delete an element from a stack (where the stack is implemented using a single linked list). Specify its time complexity using Θ notation.

(3+4+3)

4 a) Specify the postorder sequence for the binary tree whose inorder and preorder sequences are as follows:-

Inorder: B C A E D G H F I

Preorder: A B C D E F G H I

b) Write an algorithm (using C pseudocode) to print the data present in all the nodes of a binary tree (having n nodes), level by level. Within a level the data (present in the nodes) are to be printed from the left node to the right node. What is the time complexity of your algorithm?

c) Write recursive algorithms (using C pseudocode) for inorder and postorder traversal of a binary tree (where the tree is implemented using array) and specify their space and time complexities.

(4+3+3)

5 a) Write an algorithm (using C pseudocode) to reverse the direction of the links of a singly linked list during traversal.

b) What is the maximum number of items in a B-tree of order m and height h .

c) Construct a B-tree of order 5 with the keys arriving in the following sequence: 3, 7, 9, 23, 45, 1, 5, 15, 30, 27, 13, 10, 8, 19, 4, 31, 36, 55, 2, 6, 11

Then delete the keys in the following sequence: 4, 5, 9, 3, 15 and show the resulting B-tree.

(5+1+4)

6 a) If $f(n) = a_m n^m + \dots + a_1 n + a_0$ where $a_0, a_1, \dots, a_m$ are constants then prove that $f(n) = \Theta(n^m)$ for $n \geq 1$ and $a_m > 0$.

b) Write an algorithm (using C pseudocode) to search a key in a Binary Search Tree (BST) and delete it such that the resulting tree is a BST. Can the same algorithm be used for deleting an element in an AVL tree such that the resulting tree is an AVL tree? Specify the worst case time complexity to search a key in a BST and in an AVL tree using Θ notation.

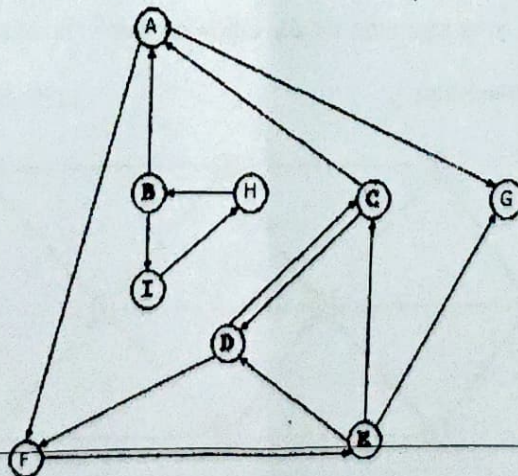
c) Explain the advantages and disadvantages of using AVL trees. Explain why the time complexity analysis of our algorithm using asymptotic notations may fail when we use disk storage.

(3+5+2)

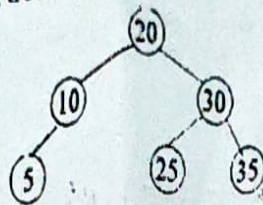
7 a) What is the output for Breadth First Traversal of the following graph if:

i) the traversal starts from vertex A

ii) the traversal starts from vertex I

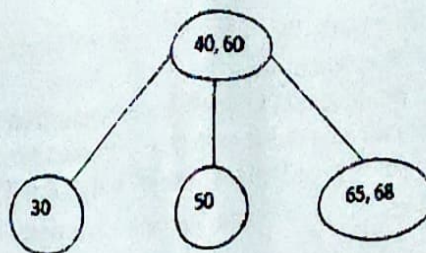


- b) Write non-recursive algorithms (using C pseudocode) for the inorder and preorder traversal of a binary tree having n nodes and specify their best case and worst case time complexities using Θ notation.
- c) Insert the keys in the tree (given below) arriving in the following sequence: 40, 45, 34 and then delete key 20 from it such that the resulting tree is an AVL tree.

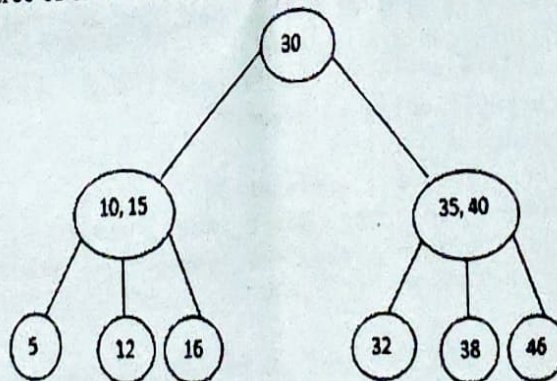


(3+5+2)

- 8 a) Insert the key 62 into the following 2-3 tree such that the resulting tree is a 2-3 tree.



- b) From the following B-tree of order 3 delete the key 30 such that the resulting tree is a B-tree



- c) Write the algorithm (using C pseudocode) for Breadth First Traversal of a graph (containing n number of vertices and e number of edges) represented using adjacency list. What is the time complexity of your algorithm?

(2+2+6)

- 9 a) Write two algorithms (using pseudocode), one should be a non-recursive algorithm for inorder traversal of threaded binary tree and the other should be an algorithm for inorder traversal of non-threaded binary tree. Compare and analyze their space and time complexities.
- b) Prove that the maximum number of nodes on level i of a binary tree is 2^{i-1} for $i \geq 1$ and the maximum number of nodes in a binary tree of depth k is $2^k - 1$ for $k \geq 1$.

- c) For the following algorithm (*tower*), specify its time and space complexity using Θ notation

Algorithm *tower*(*n*, *source*, *inter*, *dest*)

```
{
    if(n>0)
    {
        tower(n-1, source, dest, inter);
        print("Move disk", n, "from", source, "to", dest);
        tower(n-1, inter, source, dest);
    }
}
```

(6+2+2)

- 10 a) Write an algorithm which swaps the left and right children of every node of a binary tree represented using linked representation.
- b) Write algorithms (using C pseudocode) to delete an element from a Binary Search Tree (having *n* nodes) and specify its best case and worst case time complexity using Θ notation.
- c) For the following function [where, *end_index* - *start_index* = *n*-1], the worst case time complexity of *foo* (using Θ notation) is _____ and the best case time complexity (using Θ notation) of *foo* is _____ and the worst case recursion stack space required by *foo* (using Θ notation) is _____

```
int foo(int array[], int start_index, int end_index, int element){
    if (end_index >= start_index){
        int middle = start_index + (end_index - start_index) / 2;
        if (array[middle] == element)
            return middle;
        if (array[middle] > element)
            return foo(array, start_index, middle-1, element);
        return foo(array, middle+1, end_index, element);
    }
    return -1;
}
```

(4+4.5+1.5)

Attempt any five questions (including subparts)

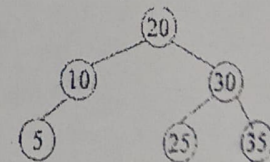
- 1 a) Obtain an addressing formula for the element $A(i_1, i_2, \dots, i_n)$ in an array declared as $A(l_1:u_1, l_2:u_2, \dots, l_n:u_n)$. Assume a sequential column major representation of the array with m bytes per element and a as the address of $A(l_1+1, l_2+2, l_3+3, \dots, l_n+n)$ where $l_k+k \leq u_k, 1 \leq k \leq n$.
 b) Write an algorithm (using C pseudocode) to implement a priority queue where the element having the minimum value is given the maximum priority.
 c) Write recursive algorithms (using C pseudocode) for inorder and postorder traversal of a binary tree (where the tree is implemented using array) and specify their space and time complexities. (3+4+3)

- 2 a) Write algorithm(s) (using C pseudocode) to insert an element into a circular-queue where the circular-queue is implemented using a doubly linked list. Specify its time complexity using Θ notation.
 b) State whether the following are true or false.
 i) $7*2^n + n^2 = O(2^n/6)$ ii) $5*2^n + n^2 = \Omega(n^{10000})$
 iii) $6n^2+4n+2 = \Theta(2^n)$ iv) $7n^2+4n+2 = O(n^{2.5})$
 v) $n/250000 = \Omega(500000)$ vi) $500005 = \Theta(5)$
 c) Write an algorithm (using C pseudocode) to reverse the direction of the links of a singly linked list during traversal. (2+3+5)

- 3 a) Specify the postorder sequence for the binary tree whose inorder and preorder sequences are as follows:-
 Inorder: B C A E D G H F I Preorder: A B C D E F G H I
 b) Write an algorithm (using C pseudocode) to print the data present in all the nodes of a binary tree (having n nodes), level by level. Within a level the data (present in the nodes) are to be printed from the left node to the right node. What is the time complexity of your algorithm?
 c) Prove that for any nonempty binary tree, T , if n_o is the number of terminal nodes and n_2 is the number of nodes of degree 2, then prove that $n_o = n_2 + 1$. (3+4+3)

- 4 a) If $f(n) = a_m n^m + \dots + a_1 n + a_0$ where $a_0, a_1, \dots, a_m$ are constants then prove that $f(n) = \Theta(n^m)$, for $n \geq 1$ and $a_m > 0$.
 b) Write an algorithm (using C pseudocode) to search a key in a Binary Search Tree (BST) and delete it such that the resulting tree is a BST. Can the same algorithm be used for deleting an element in an AVL tree such that the resulting tree is an AVL tree? Specify the worst case time complexity to search a key in a BST and in an AVL tree using Θ notation.
 c) Explain the advantages and disadvantages of using AVL trees. Explain why the time complexity analysis of our algorithm using asymptotic notations may fail when we use disk storage. (3+5+2)

- 5 a) Write an algorithm (using C pseudocode) to delete an element from a stack (where the stack is implemented using a single linked list). Specify its time complexity using Θ -notation.
 b) Write non-recursive algorithms (using C pseudocode) for the inorder and preorder traversal of a binary tree having n nodes and specify their best case and worst case time complexities using Θ notation.
 c) Insert the keys in the tree (on the right) arriving in the following sequence: 41, 50, 32 and then delete key 20 from it such that the resulting tree is an AVL tree. (3+5+2)



contains the total number of nonzero elements of the matrix. Let $X(i,0)$, $X(i,1)$ and $X(i,2)$ contain row number, column number and the value of the nonzero element in the matrix respectively, for $1 \leq i \leq t$. Write an algorithm (using C pseudocode) to store two sparse matrices (taken as input from the user) in the form (as specified above using X), perform multiplication of these matrices, store the result in the form (as specified above using X) and display the result (in the form which is usually understood by an user unfamiliar with the representation specified using X). Your algorithm should also be able to notify the user if multiplication is not possible. Also specify the time complexity of your algorithm.

b) Prove that the maximum number of nodes on level i of a binary tree is 2^{i-1} for $i \geq 1$ and the maximum number of nodes in a binary tree of depth k is $2^k - 1$ for $k \geq 1$.

c) For the algorithm *tower* (specified on the right), specify its time and space complexity using Θ notation

```

Algorithm tower(n, source, inter, dest)
{
    if(n>0)
    {
        tower(n-1, source, dest, inter);
        print("Move disk", n, "from", source, "to", dest);
        tower(n-1, inter, source, dest);
    }
}

```

(6+2+2)

- 7 a) Write an algorithm which swaps the left and right children of every node of a binary tree represented using linked representation.
- b) Write algorithms (using C pseudocode) to delete an element from a Binary Search Tree (having n nodes) and specify its best case and worst case time complexity using Θ notation.
- c) Write a tail recursive algorithm (using C pseudocode) which returns the n^{th} term of the Fibonacci series and specify its time and space complexities using Θ notation.

(4+3+3)

- 8 a) Write your recursive algorithms (using C pseudocode) to find the Greatest Common Divisor (GCD) and Least Common Multiple (LCM) of two given numbers (provided by the user as input). Specify their time complexities
- b) Write algorithms (using C pseudocode) for a menu driven implementation for performing insertion and deletion into a queue (where the queue is implemented using doubly linked list) based on user's choice and find its worst case time complexity using Θ notation.

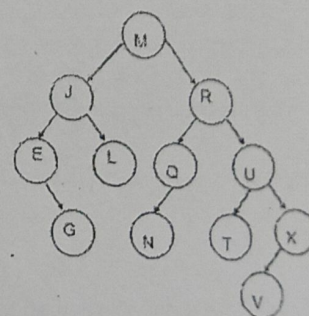
c) For the following, find the function $g(x)$ for which $f(x) = \Theta(g(x))$

i) $f(x) = 2+6x+18x^2/2! + 54x^3/3! \dots + \infty$

ii) $f(x) = 1+x+x^2/2! + x^3/3! + \dots + x^n/n!$

(4+4+2)

- 9 a) Delete L from this tree (on the right) such that the resulting tree is an AVL tree.



- b) Write an algorithm (using C pseudocode) to perform the addition of two polynomials [where each polynomial is stored using a doubly linked list of elements in the form $(m, e_{m-1}, b_{m-1}, e_{m-2}, b_{m-2}, \dots, e_0, b_0)$ where, the first entry m represents the number of non-zero terms followed by two entries (e_i, b_i) for each non-zero term (representing an exponent-coefficient pair) in the order $e_{m-1} > e_{m-2} > \dots > e_0 > 0$] and find the time complexity of your algorithm.
- c) Write an algorithm (using C pseudocode) that checks using stacks if a string of parentheses made up of $(,), \{, \}, [, \text{ and }]$ is balanced or not. For example: " $((()) g)$ " is balanced, but " $([)]$ " is not balanced since the parentheses don't match up correctly. The algorithm should return 1 if the input string is balanced and 0 otherwise.

(2+5+3)
